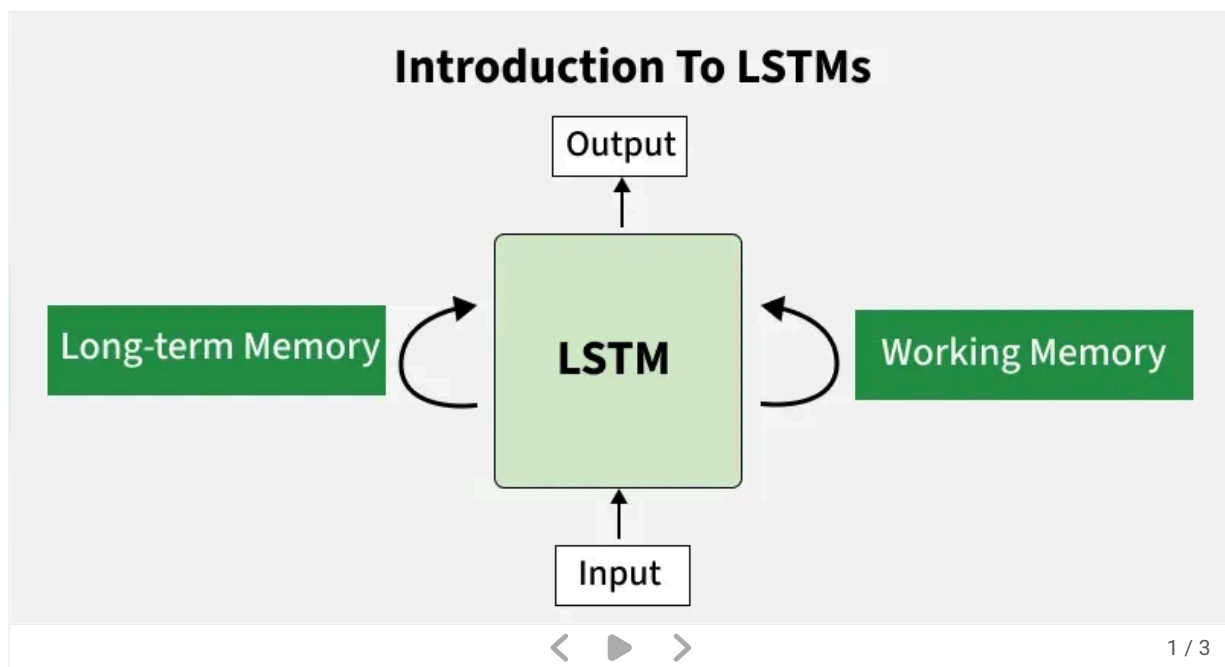


Introduction to Long Short Term Memory

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Long Short-Term Memory (LSTM) is an improved version of the [Recurrent Neural Network \(RNN\)](#) designed to capture long-term dependencies in sequential data. It uses a memory cell to store information over time, solving the limitations of traditional RNNs.

- **Handles Long Term Dependencies:** Remembers information for longer sequences
- **Memory Cell:** Stores and updates important information over time
- **Better than RNN:** Overcomes short term memory limitations
- **Applications:** Used in language translation, speech recognition and time series forecasting



Problem with Long-Term Dependencies in RNN

RNNs are designed to handle sequential data by using a hidden state that stores information from previous steps. However, they struggle to learn long-term dependencies. This happens due to:

- **Vanishing Gradient:** When training a model over time, the gradients which help the model learn can shrink as they pass through many steps. This makes it hard for the model to learn long-term patterns since earlier information becomes almost irrelevant.
- **Exploding Gradient:** Sometimes gradients can grow too large causing instability. This makes it difficult for the model to learn properly as the updates to the model become erratic and unpredictable.

LSTM Architecture

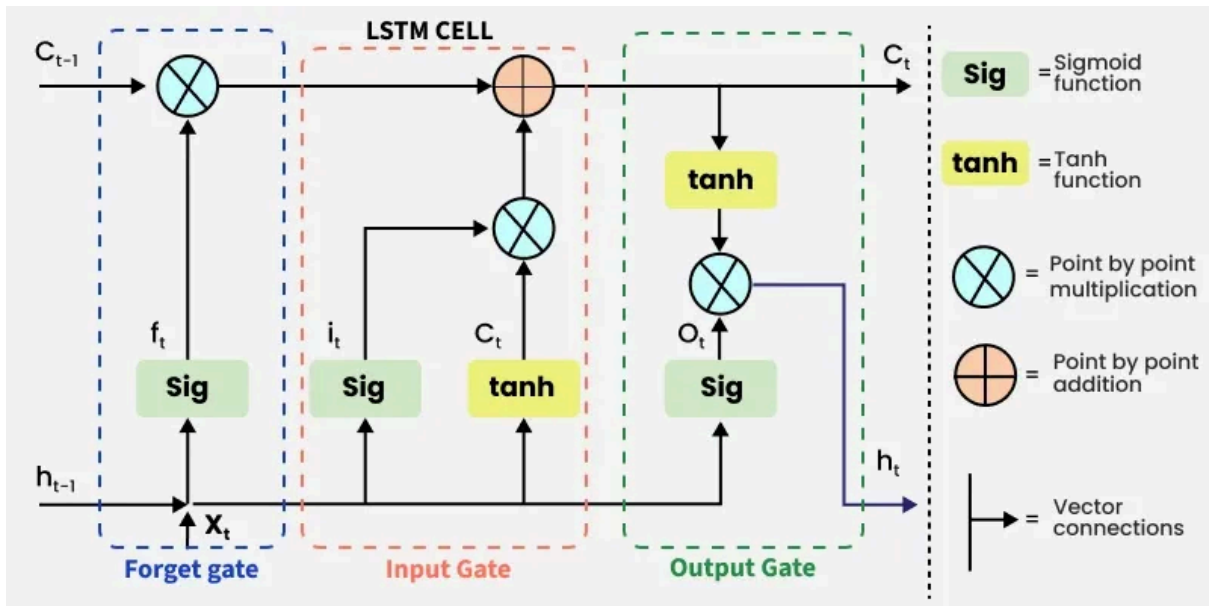
LSTM (Long Short-Term Memory) architecture is designed to learn long-term dependencies in sequential data using memory cells and gates that control the flow of information through the network.

Main Gates in LSTM

1. **Input Gate:** Decides which new information should be added to the memory cell
2. **Forget Gate:** Determines which information should be removed from the memory cell
3. **Output Gate:** Controls which information from the memory cell is passed to the next hidden state and output

Working of LSTM

LSTM consists of a repeating chain like structure with memory cells and gating mechanisms



LSTM Model

Information is retained by the cells and the memory manipulations are done by the gates. There are three gates:

1. Forget Gate

The forget gate decides which information should be kept or removed from the cell state. It uses the current input x_t and previous hidden state h_{t-1} then applies a sigmoid function to generate values between 0 and 1.

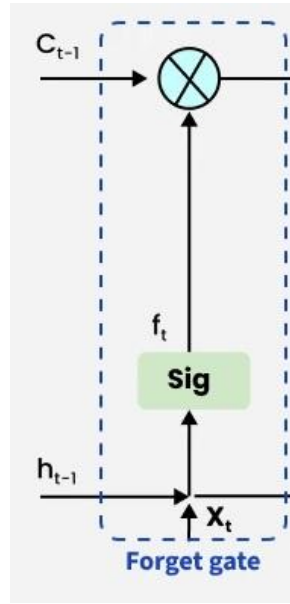
- Values close to 0 remove information
- Values close to 1 retain information
- Helps discard unnecessary past information
- Controls memory retention in the LSTM

The equation for the forget gate is:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Where

- W_f represents the weight matrix associated with the forget gate.
- $[h_t - 1, x_t]$ denotes the concatenation of the current input and the previous hidden state.
- b_f is the bias with the forget gate.
- σ is the sigmoid activation function.



2. Input gate

The addition of useful information to the cell state is done by the input gate.

- First the information is regulated using the sigmoid function and filter the values to be remembered similar to the forget gate using inputs h_{t-1} and x_t .
- Then, a vector is created using $\tanh$ function that gives an output from -1 to +1 which contains all the possible values from h_{t-1} and x_t .
- At last the values of the vector and the regulated values are multiplied to obtain the useful information.

The equation for the input gate is:

$$i_t = \sigma (W_i \cdot [h_{t-1}, x_t] + b_i)$$

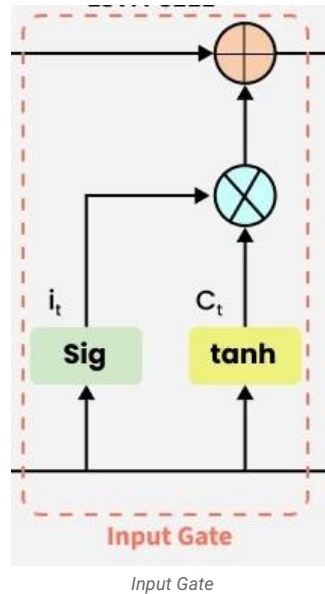
$$\hat{C}_t = \tanh (W_c \cdot [h_{t-1}, x_t] + b_c)$$

We multiply the previous state by f_t effectively filtering out the information we had decided to ignore earlier. Then we add $i_t \odot C_t$ which represents the new candidate values scaled by how much we decided to update each state value.

$$C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t$$

where

- $\odot$ denotes element-wise multiplication
- $\tanh$ is activation function



3. Output gate

The output gate determines which information from the current cell state should be passed as the hidden state (output) at the current time step. It uses the previous hidden state h_{t-1} and the current input x_t , followed by a sigmoid function to control the output flow.

$$o_t = \sigma (W_o \cdot [h_{t-1}, x_t] + b_o)$$

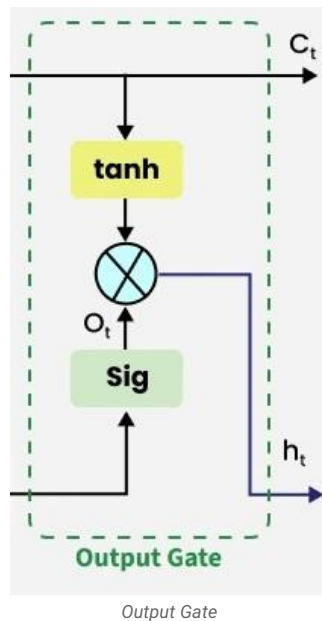
Next, the current cell state C_t is passed through a $\tanh$ activation to scale its values between -1 and $+1$. Finally, this transformed cell state is multiplied element-wise with o_t to produce the hidden state h_t :

$$h_t = o_t \odot \tanh(C_t)$$

Here:

- o_t is the output gate activation.
- C_t is the current cell state.
- $\odot$ represents element-wise multiplication.
- σ is the sigmoid activation function.

This hidden state h_t is then passed to the next time step and can also be used for generating the output of the network.



Applications

- Language modeling for machine translation and text summarization
- Speech recognition for converting audio into text
- Time series forecasting for stock prices, weather and energy usage
- Anomaly detection for fraud and intrusion detection
- Recommender systems for personalized suggestions
- Video analysis for activity recognition and motion understanding

Suggested Quiz	🔄 4 Questions
What is the main purpose of using LSTM networks over traditional RNNs? ☐ (A) LSTMs require fewer computations during training ☐ (B) LSTMs retain information across longer sequences ☐ (C) LSTMs use a simpler network architecture — Show More	
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